COMP233 — Discrete Mathematics

Lecture 5.1

Sequences

Indexed patterns • Summations • Products • Loops

$a_1, a_2, a_3, \ldots$

Σ / Π

pattern → formula → loop

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Where this lecture fits

Chapter 5 sequence

5.1 Sequences
5.2 Mathematical Induction I
5.3 Mathematical Induction II

Today's central idea

Represent indexed patterns precisely, manipulate repeated sums/products, and connect the notation to loops.

Why it matters

These tools become the language for induction, algorithm analysis, recurrence, and counting.



Learning outcomes

- Compute terms from an explicit formula and infer a plausible formula from a pattern.
- Translate between expanded sums/products and Σ/Π notation.
- Use telescoping, algebraic properties, factorials, and change of variable correctly.
- Translate mathematical notation to equivalent loop logic and explain indexing.

Motivation: discrete patterns are everywhere

Growth pattern

Ancestors by generation:
2, 4, 8, 16, 32, ...

$A_k = 2^k$

Scheduling pattern

Departure times, sampled signals, packet counts, and iteration states are naturally indexed: t_0, t_1, t_2, ...

Programming pattern

A loop generates or accumulates a sequence one index at a time. Indexing errors often become program errors.

Predict before formalizing

$$2, 4, 8, 16, \ldots \rightarrow ? \rightarrow a_k = ?$$

$$1, -1, 1, -1, \ldots \rightarrow c_j = ?$$

Think like a mathematician

A pattern suggests a rule. A rule defines terms. Later, induction will prove formulas that claim to work for every index.

What is a sequence?

Definition (precise)

A sequence is a function whose domain is either all integers between two given integers, or all integers greater than or equal to a given integer.

How we write it

$a_m, a_{m+1}, a_{m+2}, \ldots, a_n$

- a_k is a term
- k is the index (subscript)
- m is the initial index

$$k \mapsto a_k$$

Order matters

(1, 2, 3) and (3, 2, 1) contain the same values but are different sequences.

Important correction

Do not think of a sequence as an unordered set. The index is part of the mathematical object.

Example

If $a_k = 2^k$ for $k \geq 1$, then $a_1=2$, $a_2=4$, $a_3=8$, … . The starting index is part of the definition.

Explicit formulas: compute terms directly

Example A

$a_k = k/(k+1),\ k \geq 1$

$a_1=1/2,\ a_2=2/3,\ a_3=3/4,\ a_4=4/5,\ a_5=5/6$

Example B

$b_i = (i-1)/i,\ i \geq 2$

$b_2=1/2,\ b_3=2/3,\ b_4=3/4,\ b_5=4/5,\ b_6=5/6$

Observation

These two sequences list the same numerical values but use different index rules and starting indices. Read the domain before substituting.

Concept check

What is the first legal input for b_i? Why is b_1 not part of this sequence?

Indexing can start at 0 — and it changes the formula

Alternating sequence

$c_j = (-1)^j,\ j \geq 0$

$c_0=1,\ c_1=-1,\ c_2=1,\ c_3=-1,\ c_4=1,\ c_5=-1$

Same pattern, different indexing

$d_k = (-1)^{k+1},\ k \geq 1$

$d_1=1,\ d_2=-1,\ d_3=1,\ d_4=-1,\ \ldots$

Boundary case

The starting index affects both the first term and the algebraic form. Always record the index condition together with the formula.

Quick check

Would $(-1)^k$ generate $1, -1, 1, -1, \ldots$ when k starts at 1? Explain.

From initial terms to an explicit formula

$$1,\ -1/4,\ 1/9,\ -1/16,\ 1/25,\ -1/36,\ \ldots$$

Step 1 — magnitude

Denominators are squares: $1^2, 2^2, 3^2, 4^2, \ldots$
So magnitude is $1/k^2$.

Step 2 — sign

Signs alternate $+, -, +, -, \ldots$
Starting positive at $k=1$ suggests $(-1)^{k+1}$.

Step 3 — combine

$$a_k = (-1)^{k+1}/k^2,\ k \geq 1$$

Equivalent indexing

$a_k = (-1)^k/(k+1)^2$, $k \geq 0$, generates the same listed values with the index shifted by one.

Caution

A finite list may fit many formulas. Pattern recognition proposes a rule; it does not prove a unique rule.

Concept check: index, term, formula

A

For $a_k = 3k-2$, $k \geq 1$:
Find a_1, a_2, a_5.

B

For $b_j = (-1)^j/(j+1)$, $j \geq 0$:
Write the first four terms.

C

Give one explicit formula for:
2, 5, 8, 11, ...

Think first. Then compare your reasoning:

A

1, 4, 13

B

$1, -1/2, 1/3, -1/4$

C

$a_k = 3k-1$, $k \geq 1$

Summation notation (Σ): repeated addition

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n$$

Σ

summation operator (repeated addition)

k

index / dummy variable

m

lower limit (starting index)

n

upper limit (ending index)

How many terms?

From k=m through k=n inclusive, there are n−m+1 terms.

Read aloud

“The sum from k equals m to n of a-sub-k.”

Computing summations

Given sequence values

$a_1=-2, a_2=-1, a_3=0, a_4=1, a_5=2$

$$\Sigma^{5}_{k=1}\, a_k = -2 -1 +0 +1 +2 = 0$$

Given a formula

Compute $\Sigma^{5}_{k=1}\, k^2$

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

Single-term sum

$$\Sigma^{2}_{k=2}\, a_k = a_2 = -1$$

Indexed term

$$\Sigma^{2}_{k=1}\, a_{2k} = a_2 + a_4 = -1 + 1 = 0$$

Strategy

When unsure, expand the sum. Expansion exposes the actual indices being used.

Two essential translations: expand ↔ compress

Σ → expanded form

$$\sum_{i=0}^{n} (-1)^i/(i+1)$$
$$= 1 - 1/2 + 1/3 - 1/4 + \cdots + (-1)^n/(n+1)$$

Expanded form → Σ

$$1/n + 2/(n+1) + 3/(n+2) + \cdots + (n+1)/(2n)$$
$$= \sum_{k=0}^{n} (k+1)/(n+k)$$

Compression checklist

1) Identify the changing quantity. 2) Find its start/end indices. 3) Express every term with the same index. 4) Verify the first and last terms.

Common error

Do not guess the upper limit from the number of symbols you see. Check the first and last generated terms.

Separating off — or adding on — a final term

Separate the final term

$$\sum_{i=1}^{n+1} 1/i^2$$
$$= \sum_{i=1}^{n} 1/i^2 + 1/(n+1)^2$$

Add a final term

$$\sum_{k=0}^{n} 2^k + 2^{n+1}$$
$$= \sum_{k=0}^{n+1} 2^k$$

Expose last term → Apply a rule → Simplify → Recombine

Why this matters

This move appears constantly in induction: P(k+1) is often rewritten as “the first k terms + the new final term.”

Telescoping sums: cancellation reveals the structure

$\Sigma^{n}_{k=1}\ 1/[k(k+1)]$ $\quad 1/[k(k+1)] = 1/k - 1/(k+1)$ **expand**

$$(1-1/2) + (1/2-1/3) + \cdots + (1/n-1/(n+1))$$

Cancellation

Interior terms occur once positively and once negatively, so they cancel in pairs.

Result = 1 − 1/(n+1)

Important

A telescoping sum is not “random cancellation.” Expand enough terms to justify exactly which terms cancel.

Product notation (Π): repeated multiplication

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \cdots \cdot a_n$$

Example

$$\prod_{k=1}^{5} k = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$$

Single-term product

$$\prod_{k=1}^{1} k/(k+1) = 1/2$$

Parallel idea

Σ combines terms by addition; Π combines terms by multiplication. Their algebraic properties are similar in some ways—but not identical.

Quick check

$\prod_{k=1}^{4} (k+1) = 2 \cdot 3 \cdot 4 \cdot 5 = 120 = 5!$, not $4!$.

Factorial notation

Definition

For $n \geq 1$:
$n! = n(n-1)(n-2)\cdots 2\cdot 1$

Boundary case: $0! = 1$

Product form

$$n! = \prod_{k=1}^{n} k$$

Recursive form

$0! = 1$
$n! = n\cdot(n-1)!$ for $n \geq 1$

Examples

$4! = 24$
$6! = 720$
$9! = 362{,}880$

Why 0! = 1?

It makes the recursion consistent: $1! = 1\cdot 0! = 1$, so $0!$ must equal 1.

Computing with factorials: expand only what you need

Cancel common factors

$8!/7! = 8 \cdot 7!/7! = 8$

Mixed factorials

$5!/(2! \cdot 3!) = (5 \cdot 4 \cdot 3!)/(2 \cdot 1 \cdot 3!) = 10$

Symbolic ratio

$(n+1)!/n! = (n+1) \cdot n!/n! = n+1$

Useful identity

$n!/(n-3)! = n(n-1)(n-2) \quad (\text{for } n \geq 3)$

Avoid

Do not distribute factorial over addition: $(a+b)! \neq a!+b!$ in general.

Properties of summations and products

Sum of sums

$$\Sigma\, a_k + \Sigma\, b_k = \Sigma(a_k + b_k)$$

Constant multiple

$$c \cdot \Sigma\, a_k = \Sigma(c a_k)$$

Product of products

$$(\Pi\, a_k)(\Pi\, b_k) = \Pi(a_k b_k)$$

Matching limits matter

These identities pair terms index-by-index over the same lower and upper limits.

Worked combination

If $a_k = k+1$ and $b_k = k-1$, then $\Sigma a_k + 2\Sigma b_k = \Sigma[(k+1)+2(k-1)] = \Sigma(3k-1)$.

Valid transformations vs common mistakes

✓ **Valid**

$$\Sigma(a_k+b_k) = \Sigma a_k + \Sigma b_k$$

✗ **Not valid in general**

$$(\Sigma a_k)(\Sigma b_k) = \Sigma(a_k b_k)$$

✓ **Valid**

$$\Pi(a_k b_k) = (\Pi a_k)(\Pi b_k)$$

✗ **Not valid in general**

$$\Sigma(a_k b_k) = (\Sigma a_k)(\Sigma b_k)$$

Counterexample

Let $a_1=a_2=b_1=b_2=1$. Then $(\Sigma a_k)(\Sigma b_k)=2\cdot 2=4$, but $\Sigma(a_k b_k)=1+1=2$.

Rule of thumb

When uncertain, expand both sides for a two-term case.

Dummy variable vs genuine change of variable

Rename only — same sum

$\Sigma^{3}_{k=1}\, k^2 = \Sigma^{3}_{i=1}\, i^2$

The index name is bound locally; changing the symbol alone changes nothing.

Re-index — same values, new bounds

$\Sigma^{4}_{j=2}\,(j-1)^2 = 1^2+2^2+3^2 = \Sigma^{3}_{k=1}\, k^2$

Key distinction

A dummy-variable rename keeps the bounds and term structure identical. A change of variable transforms the bounds and the summand together.

Not equivalent

$\Sigma^{n}_{k=1}\, k^2 \neq \Sigma^{n+1}_{k=2}\, k^2$ in general: the index runs through different values.

Change of variable: transform bounds and the term together

Start

$\sum_{k=0}^{6} 1/(k+1)$

Let

$j = k+1 \Rightarrow k = j-1$

New bounds

$k=0 \rightarrow j=1$
$k=6 \rightarrow j=7$

Change bounds → Rewrite term → Rename if desired → Verify

$$\sum_{k=0}^{6} 1/(k+1) = \sum_{j=1}^{7} 1/j$$

Verification habit

Check the first term and the last term after re-indexing. If both match, the transformation is usually on track.

Σ and Π are loop-shaped mathematics

Mathematical sum

$$S = \sum_{k=1}^{n} k^2$$

```
S = 0
for k = 1 to n
    S = S + k*k
```

Mathematical product

$$P = \prod_{k=1}^{n} k$$

```
P = 1
for k = 1 to n
    P = P * k
```

Why different initial values?

0 is the additive identity; 1 is the multiplicative identity.

Equivalent loops can use different indices

Loop A

```
for i = 1 to n
    print a[i]
```

Loop B

```
for j = 0 to n−1
    print a[j+1]
```

Loop C

```
for k = 2 to n+1
    print a[k−1]
```

All three visit $a_1, a_2, \ldots, a_n$ in the same order.

Equivalent accumulation

s := a[1]; for k=2..n: s:=s+a[k]

Same final result as starting s:=0 and looping k=1..n.

Index-shift danger

Equivalent code requires both the loop range and the array expression to shift consistently.

Connection

This is the programming version of change of variable in a summation.

Applications: sequences as computational traces

Computer Science

Running time often counts operations across indexed iterations. A loop with k=1..n naturally produces a sequence of states or costs.

Cybersecurity

Key-space growth can be modeled by sequences such as 10^k or 2^k, making exponential growth visible and comparable.

AI / Data

Iterative optimization, training epochs, and sampled measurements are indexed sequences; sums aggregate loss, error, or counts.

Transfer example

A password system with k decimal digits has $a_k = 10^k$ possible strings (repetition allowed). Compare a_4=10,000 with a_8=100,000,000.

Lesson

The same notation supports mathematical reasoning and computational implementation.

Bridge to Lecture 5.2: pattern is not proof

$$1 + 2 + 3 + \cdots + n = n(n+1)/2$$

Test a few cases

n=1 ✓
n=2 ✓
n=3 ✓
n=4 ✓

But...

Checking finitely many examples cannot prove a statement for every positive integer.

Next lecture

Mathematical induction proves a whole infinite family of indexed statements.

Base case → Assume P(k) → Prove P(k+1) → Conclude all n

Lecture synthesis: one connected language

Pattern → Sequence a_k → Σ / Π → Loop → Induction next

Remember

- A sequence is indexed.
- Explicit formulas compute terms.
- Σ and Π compress repeated operations.
- Indices may be renamed or re-indexed carefully.

Use correctly

- Expand when uncertain.
- Check first/last terms after re-indexing.
- Use telescoping only when cancellation is justified.
- Respect matching limits in algebraic properties.

Connect to code

- Σ → initialize 0 and add.
- Π → initialize 1 and multiply.
- Loop variable ↔ summation index.
- Equivalent loops require consistent index shifts.

Key sentence

Sequences organize indexed patterns; Σ and Π compress repeated operations; loops execute those same patterns.

Core concepts

Sequence

Function on an integer index domain; order matters.

Explicit formula

Direct rule for a_k in terms of k.

Σ / Π

Compact notation for repeated addition / multiplication.

Telescoping

Expanded neighboring terms cancel systematically.

Factorial

$n! = \Pi^{n}_{k=1}\ k$, with $0!=1$.

Dummy variable

Bound index name can change without changing the sum.

Change of variable

Transform index, bounds, and summand consistently.

Loop connection

Accumulator mirrors Σ or Π term by term.

Induction bridge

Sequence formulas motivate statements P(n) for all n.

Common misconceptions

Sequence ≠ unordered set

Index/order is essential.

Upper limit ≠ number of terms

$\Sigma^{9}_{k=4}$ has $9-4+1=6$ terms.

Rename ≠ re-index

Changing $k \to i$ is trivial; changing bounds needs a transformed term.

Σ rules ≠ Π rules

Do not transfer identities blindly.

Telescoping ≠ arbitrary cancellation

Only equal opposite terms cancel.

Pattern ≠ proof

Examples can suggest a formula, not prove it for all n.

Diagnostic habit

Expand a short case whenever a symbolic manipulation feels uncertain.

Conceptual connections

Idea	Mathematical meaning	Programming analogue
a_k	value at index k	array/state at iteration k
Σ	repeated addition	accumulator initialized at 0
Π	repeated multiplication	accumulator initialized at 1
dummy variable	bound index name	loop-variable name
change of variable	re-index with transformed bounds	shift loop range + expression
separate final term	first k terms + new term	loop state before + next iteration

Worked cases: valid, invalid, boundary

Valid

$\Sigma^{4}_{k=1}\,(2k-1)$
$= 1+3+5+7 = 16$

Invalid claim

$(\Sigma^{4}_{k=1}\,k)^2 = \Sigma^{4}_{k=1}\,k^2$
$100 \neq 30$

Boundary

$\Sigma^{3}_{k=3}\,a_k = a_3$
One legal index → one term.

Factorial boundary

$0! = 1$
$1! = 1$

Telescoping

$\Sigma^{3}_{k=1}\,(1/k-1/(k+1))$
$= 1-1/4 = 3/4$

Re-index

$\Sigma^{3}_{k=0}\,1/(k+1) = \Sigma^{4}_{j=1}\,1/j$

Transfer

For every transformation, ask: what values does the index actually visit?

Question 1 / 8

QUESTION

Which statement is the best formal description of a sequence?

A. An unordered set of numbers

B. A function defined on an integer index domain

C. Any finite list

D. A sum of terms

Question 2 / 8

QUESTION

For $a_k = (-1)^{k+1}/k^2$, $k \geq 1$, what is a_4?

A. 1/16

B. −1/16

C. −1/4

D. 1/4

Question 3 / 8

QUESTION

How many terms are in $\Sigma^{9}_{k=4}\, a_k$?

A. 5

B. 6

C. 9

D. 13

Question 4 / 8

QUESTION

Which expression equals $1 + 3 + 5 + \cdots + (2n-1)$?

A. $\sum_{k=1}^{n} (2k-1)$

B. $\sum_{k=0}^{n} (2k-1)$

C. $\sum_{k=1}^{n} (2k+1)$

D. $\sum_{k=0}^{n} 2k$

Question 5 / 8

QUESTION

Which identity is valid when the sums have the same limits?

A. $(\Sigma a_k)(\Sigma b_k)=\Sigma(a_k b_k)$

B. $\Sigma(a_k+b_k)=\Sigma a_k+\Sigma b_k$

C. $\Sigma(a_k b_k)=(\Sigma a_k)(\Sigma b_k)$

D. $\Sigma a_k^2=(\Sigma a_k)^2$

Question 6 / 8

QUESTION

After j = k+1, what does $\Sigma^{6}_{k=0}\ 1/(k+1)$ become?

A. $\Sigma^{6}_{j=0}\ 1/j$

B. $\Sigma^{7}_{j=1}\ 1/j$

C. $\Sigma^{6}_{j=1}\ 1/(j+1)$

D. $\Sigma^{7}_{j=0}\ 1/(j+1)$

Question 7 / 8

QUESTION

True or false: checking a formula for n=1,2,3,4 proves it for all positive integers.

A. True

B. False

Question 8 / 8

QUESTION

Which loop initialization matches a product $\Pi^{n}_{k=1} a_k$?

A. p = 0

B. p = 1

C. p = n

D. p = a[n] only

Answers with reasoning — 1 to 4

Q1 — B

A sequence is a function on an integer index domain; a displayed list is a representation, not the definition.

Q2 — B

$a_4=(-1)^5/4^2=-1/16$. The exponent controls sign; the denominator is 4^2.

Q3 — B

Indices 4,5,6,7,8,9 give $9-4+1=6$ terms.

Q4 — A

At k=1 the term is 1; at k=n the term is $2n-1$. The other choices start or end incorrectly.

Common errors

Q2: forgetting the +1 in the exponent. Q3: using upper−lower instead of upper−lower+1. Q4: failing to verify both first and last terms.

Answers with reasoning — 5 to 8

Q5 — B

Summation is linear: termwise addition can be split or combined. A product of sums contains cross-terms.

Q6 — B

j=k+1 changes the bounds 0→1 and 6→7, while 1/(k+1) becomes 1/j.

Q7 — False

Finite testing gives evidence, not a universal proof. Mathematical induction is designed for such claims.

Q8 — B

Products accumulate from the multiplicative identity 1; starting from 0 would force every product to 0.

Common errors

Q5: treating Σ as if it distributed over multiplication like ordinary factoring. Q6: changing the variable without changing bounds. Q8: copying the sum pattern without considering the identity element.

Optional enrichment: sequences inside algorithms

Repeated division

Binary conversion repeatedly divides an integer by 2. Quotients form a decreasing sequence; remainders encode binary digits.

Growth comparison

Linear: $a_k = k$
Polynomial: $a_k = k^2$
Exponential: $a_k = 2^k$
Their growth rates quickly diverge.

Security intuition

A key space that grows like 2^k becomes dramatically larger when k increases—a concrete reason sequence growth matters.

Mini-example

Start with 13: 13 div 2=6 rem 1 → 6 div 2=3 rem 0 → 3 div 2=1 rem 1 → 1 div 2=0 rem 1. Read remainders backward: 1101_2.

Connection

An algorithm often generates a sequence of states; discrete mathematics gives us notation to describe and reason about that sequence.

References & course resources

Primary text

Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., Chapter 5, Section 5.1.

Original course materials

Prof. Mustafa Jarrar — www.jarrar.info/courses/DMath/

Updated course resources

COMP233 GitHub — RuwaYafa/BZU_Courses/COMP233

With appreciation for subsequent course-material contributions, including Dr. Radi Jarrar.